

ENM 5310 — Quiz 2 · VERSION A

10 minutes. Closed book, closed notes, no devices. One best answer per question. On your bubble sheet: write and bubble your name and Penn ID, and bubble **Version A**. Pencil only. Each question has five bubbles (A–E) — ignore E. No penalty for wrong answers, so answer everything.

- The data distribution p^* is bimodal, with two well-separated modes. You fit a single Gaussian by maximum likelihood using a very large dataset. Which statement best describes the result?
 - It converges to whichever mode carries more probability mass, since maximum likelihood is mode-seeking.
 - It converges to the minimizer of $\text{KL}(p_\theta \| p^*)$, which is why it locks onto a single mode.
 - It converges to the minimizer of $\text{KL}(p^* \| p_\theta)$; because that divergence heavily penalizes putting low density where the data has mass, the fitted Gaussian is wide and straddles both modes, with most of its mass in the valley between them.
 - No maximizer exists once the model is misspecified, so the fit fails to converge.
- Let $\hat{w}$ minimize $\|y - \Phi w\|_2^2$, with Φ of full column rank. Which statement is correct?
 - The normal equations $\Phi^\top(y - \Phi \hat{w}) = 0$ say the residual is orthogonal to every column of Φ , so $\hat{y} = \Phi \hat{w}$ is the orthogonal projection of y onto the column space of Φ ; and $\hat{w}$ is exactly the MLE under $\epsilon \sim \mathcal{N}(0, \sigma^2 I)$, with the same $\hat{w}$ for every value of σ^2 .
 - Because the model is linear in w , it can represent only straight-line relationships between x and y ; capturing curvature requires abandoning least squares.
 - The squared-error loss is chosen because it is smooth and convex; the Gaussian noise assumption is needed only to attach error bars afterwards, not to justify the loss.
 - Least squares is an approximation to maximum likelihood which becomes exact only in the limit $N \rightarrow \infty$.
- You fit linear regression with Gaussian noise of variance σ^2 and a prior $w \sim \mathcal{N}(0, \tau^2 I)$. Which statement is correct?
 - Ridge is unbiased and has smaller variance than least squares, which is why it is preferred.
 - The MAP estimate is the ridge solution with $\lambda = \sigma^2/\tau^2$, and $\Phi^\top \Phi + \lambda I$ is invertible for every $\lambda > 0$, so the estimate exists and is unique even when $\Phi^\top \Phi$ is singular.
 - The prior matters only when $N < D$; for $N \geq D$ the MAP estimate and the MLE coincide exactly.
 - The MAP estimate is the ridge solution with $\lambda = \tau^2/\sigma^2$; the prior rescales the coefficients but cannot affect whether a solution exists.
- Compare gradient descent, $\theta \leftarrow \theta - \eta \nabla f(\theta)$, with Newton's method for minimizing a smooth f . Which statement is correct?
 - Newton's method is gradient descent with the step size chosen automatically: the two move in the same direction and differ only in how far they travel along it.
 - Because it uses curvature, Newton's method converges to a better minimum than gradient descent when the objective has several local minima.
 - Newton's step is the exact minimizer of the second-order Taylor model of f at the current iterate, obtained by solving $\nabla^2 f(\theta) \delta = -\nabla f(\theta)$; it needs no step size and is exact in one step on a quadratic, but each iteration costs $O(D^3)$ to factor the Hessian, which is why large models are trained with first-order methods.
 - Newton's method applies only when the objective is exactly quadratic; for any other objective the step is undefined.
- Consider maximum likelihood for logistic regression, $p(y = 1 | x, w) = \sigma(w^\top \phi(x))$. Which statement is correct?
 - The Hessian is $\Phi^\top \Phi$, independent of w , so a single Newton step from any starting point reaches the exact optimum.
 - The negative log-likelihood is nonconvex, so gradient descent may converge to a poor local minimum and several restarts are essential.
 - A closed-form solution does exist but is numerically unstable, which is why iterative methods are used in practice.
 - The gradient is $\Phi^\top(\mu - y)$ with $\mu = \sigma(\Phi w)$; setting it to zero leaves w inside a sigmoid, so there is no closed form — but the negative log-likelihood is convex, so any stationary point is a global minimum.
- In Bayesian linear regression the posterior predictive variance at a test input is $\sigma^2 + \phi(x^*)^\top S_N \phi(x^*)$. Which statement is correct?
 - The first term is epistemic and is reduced by collecting more data; the second is the irreducible observation noise.
 - The first term is aleatoric — independent of x^* and of N — while the second is epistemic, varies with x^* , and shrinks like $O(1/N)$; with unlimited data the predictive variance tends to σ^2 , not to zero.
 - Both terms shrink to zero as $N \rightarrow \infty$, so with enough data the predictions become exact.
 - The split is valid only at the training inputs; at a new x^* the two contributions cannot be separated.

END — confirm you bubbled Version A.

ENM 5310 — Quiz 2 · VERSION B

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- The data distribution p^* is bimodal, with two well-separated modes. You fit a single Gaussian by maximum likelihood using a very large dataset. Which statement best describes the result?
 - No maximizer exists once the model is misspecified, so the fit fails to converge.
 - It converges to the minimizer of $\text{KL}(p^* \parallel p_\theta)$; because that divergence heavily penalizes putting low density where the data has mass, the fitted Gaussian is wide and straddles both modes, with most of its mass in the valley between them.
 - It converges to the minimizer of $\text{KL}(p_\theta \parallel p^*)$, which is why it locks onto a single mode.
 - It converges to whichever mode carries more probability mass, since maximum likelihood is mode-seeking.
- Let $\hat{w}$ minimize $\|y - \Phi w\|_2^2$, with Φ of full column rank. Which statement is correct?
 - Because the model is linear in w , it can represent only straight-line relationships between x and y ; capturing curvature requires abandoning least squares.
 - The squared-error loss is chosen because it is smooth and convex; the Gaussian noise assumption is needed only to attach error bars afterwards, not to justify the loss.
 - Least squares is an approximation to maximum likelihood which becomes exact only in the limit $N \rightarrow \infty$.
 - The normal equations $\Phi^\top(y - \Phi \hat{w}) = 0$ say the residual is orthogonal to every column of Φ , so $\hat{y} = \Phi \hat{w}$ is the orthogonal projection of y onto the column space of Φ ; and $\hat{w}$ is exactly the MLE under $\epsilon \sim \mathcal{N}(0, \sigma^2 I)$, with the same $\hat{w}$ for every value of σ^2 .
- You fit linear regression with Gaussian noise of variance σ^2 and a prior $w \sim \mathcal{N}(0, \tau^2 I)$. Which statement is correct?
 - The MAP estimate is the ridge solution with $\lambda = \tau^2/\sigma^2$; the prior rescales the coefficients but cannot affect whether a solution exists.
 - The prior matters only when $N < D$; for $N \geq D$ the MAP estimate and the MLE coincide exactly.
 - The MAP estimate is the ridge solution with $\lambda = \sigma^2/\tau^2$, and $\Phi^\top \Phi + \lambda I$ is invertible for every $\lambda > 0$, so the estimate exists and is unique even when $\Phi^\top \Phi$ is singular.
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 - The first term is epistemic and is reduced by collecting more data; the second is the irreducible observation noise.
 - The first term is aleatoric — independent of x^* and of N — while the second is epistemic, varies with x^* , and shrinks like $O(1/N)$; with unlimited data the predictive variance tends to σ^2 , not to zero.

END — confirm you bubbled Version B.